ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** hypertonic	Water leaves.
2	**(d)** 8	Same number.
3	**(d)** mitochondria	Energy.
4	**(a)** growing region	Meristem.
5	**(b)** swell and may burst	Hypotonic environment.
6	**(c)** 4	Half number.
7	**(b)** chloroplast	Plants only.
8	**(a)** mitosis	Mitosis.
9	**(c)** protein synthesis	Ribosomes make proteins.
10	**(a)** isotonic	Equal concentration.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	RBC has no cell wall (1); Plant cell wall resists the expansion (1).	1+1
12	Mitosis: 12 (1); Meiosis: 6 (1).	1+1
13	Rough has ribosomes; smooth lacks (1); Rough makes proteins; smooth lipids (1).	1+1
14	More space for haemoglobin (1); Carries more oxygen (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	In water: mass increases (1); In salt solution: mass decreases (1); Osmosis through membranes (1).	1+1+1
16	Bacterium has no nucleus, only a nucleoid (1); Animal cell has a nucleus with chromosomes (1); Bacteria have cell walls (1).	1+1+1
17	Carry impulses over long distances (1); Branches receive and send signals (1); Structure suits function (1).	1+1+1
	OR: As a cell grows, volume rises faster than surface area (1); Materials cannot reach the interior fast enough (1); Many small cells exchange better (1).	1+1+1
18	Cell size is not related to the body size (1); Larger organisms have more cells (1); Cell number differs (1).	1+1+1
	OR: Organelles serve as the functions (1); Food vacuole digests (1); Contractile vacuole removes excess water (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Double membrane with pores (1); Nucleolus: ribosome synthesis (1); Chromatin: DNA and proteins (1); Control of cell activities (1); Heredity and division (1).	1+1+1+1+1
	OR: Nucleus: manager (1); Ribosomes: workers (1); ER and Golgi: transport and packing (1); Mitochondria: power (1); Lysosomes: waste disposal; membrane: gate (1).	1+1+1+1+1
20	Diffusion of O2 and CO2 (1); Osmosis of water (1); Active transport against gradient needs energy (1); Example: root cells take up minerals (1); Selective permeability (1).	1+1+1+1+1
	OR: Nucleus: absent (nucleoid) vs present (1); Membrane-bound organelles: absent vs present (1); Size: generally smaller vs larger (1); Example: bacterium vs plant or animal cell (1); Genetic material: not enclosed vs enclosed by a nuclear membrane (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that bacteria have a nucleus, or that prokaryotes have no genetic material; the DNA is present but is not enclosed in a nuclear membrane.
- Mixing up hypotonic and hypertonic: in a *hypotonic* outside solution the cell swells, in a *hypertonic* one it shrinks.
- Calling osmosis the movement of any solute; osmosis is the movement of *water* through a selectively permeable membrane.

- Confusing the functions of rough ER (protein synthesis, studded with ribosomes) and smooth ER (lipids, detoxification), or saying that mitochondria 'make' energy; they release energy from food as ATP.
- Saying that the cell wall and cell membrane are the same; the membrane is selectively permeable and present in all cells, the wall is rigid and fully permeable.
- Forgetting that plant cells do not burst in water because of the cell wall, whereas an animal cell like RBC can swell and burst.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** detoxification	Detox.
2	**(a)** nuclear membrane	Double membrane.
3	**(b)** a muscle cell	High energy demand.
4	**(b)** lose mass	Water leaves.
5	**(b)** mitochondria	Has its own DNA.
6	**(a)** exchange between nucleus and cytoplasm	Transport.
7	**(c)** a gland cell secreting proteins	Secretion.
8	**(c)** gain mass	Water enters.
9	**(a)** leaf	Photosynthetic.
10	**(d)** chromosomes	During division.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Place in cold and hot water (1); Hot water damages the membrane and pigment leaks (1).	1+1
12	Cell wall is rigid (1); It becomes turgid (1).	1+1
13	Water leaves microbes by osmosis (1); They cannot grow (1).	1+1
14	Particles have higher kinetic energy (1); Diffusion is faster (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Double membrane with folded inner membrane (cristae) (1); Large surface for reactions (1); Releases energy as ATP (1).	1+1+1
16	Dilute: cell swells, turgid (1); Isotonic: no change (1); Concentrated: plasmolysis (1).	1+1+1
17	Water moves from dilute to concentrated solution (1); Through the membrane (1); Until equal or balanced by pressure (1).	1+1+1
	OR: Proteins made on rough ER (1); Packed in vesicles to Golgi (1); Golgi modifies and sends them out (1).	1+1+1
18	Mitochondria in all eukaryotes; chloroplasts in plants (1); Mitochondria release energy; chloroplasts make food (1); Chloroplast has chlorophyll and thylakoids (1).	1+1+1
	OR: Equal size cubes (1); Wipe before weighing (1); Same time (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Chloroplasts (1); Chromoplasts (1); Leucoplasts (1); Examples in plants (1); Origin and own DNA (1).	1+1+1+1+1
	OR: Plant cell turgid (1); Animal cell swells, bursts (1); Wall prevents bursting (1); Amoeba expels water (1); Osmoregulation (1).	1+1+1+1+1
20	Nucleus (1); Wall (1); Plastids (1); Vacuoles (1); Examples (1).	1+1+1+1+1
	OR: Light microscope (1); Stains (1); Electron microscope for organelles (1); Cell culture (1); Discoveries: Hooke, Brown (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that bacteria have a nucleus, or that prokaryotes have no genetic material; the DNA is present but is not enclosed in a nuclear membrane.
- Mixing up hypotonic and hypertonic: in a *hypotonic* outside solution the cell swells, in a *hypertonic* one it shrinks.
- Calling osmosis the movement of any solute; osmosis is the movement of *water* through a selectively permeable membrane.
- Confusing the functions of rough ER (protein synthesis, studded with ribosomes) and smooth ER (lipids, detoxification), or saying that mitochondria 'make' energy; they release energy from food as ATP.

- Saying that the cell wall and cell membrane are the same; the membrane is selectively permeable and present in all cells, the wall is rigid and fully permeable.
- Forgetting that plant cells do not burst in water because of the cell wall, whereas an animal cell like RBC can swell and burst.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** red blood cells of mammals	More space for haemoglobin.
2	**(a)** the cell membrane is damaged	Heat damages the membrane.
3	**(a)** cell	Cell.
4	**(b)** many identical plants from a small piece	Cloning.
5	**(d)** animal cell	Animal cell.
6	**(b)** cells absorb water by osmosis	Turgidity restored.
7	**(d)** nerve cell	Neuron.
8	**(b)** can develop into different types of cells	Unspecialised.
9	**(d)** nucleoid	No membrane.
10	**(a)** bacteria lose water by osmosis	Plasmolysis of microbes.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Large surface area to volume ratio (1); Efficient exchange of materials (1).	1+1
12	Muscle needs a lot of energy (1); Mitochondria release energy (1).	1+1
13	Water enters by osmosis (1); Vacuole expels excess water (1).	1+1
14	They have their own DNA and ribosomes (1); They can make some of their own proteins (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Wall: cellulose, rigid, fully permeable (1); Membrane: lipid-protein, flexible, selective (1); Wall only in plants (1).	1+1+1
16	Reproduction in unicellular (1); Growth in multicellular (1); Repair of tissues (1).	1+1+1
17	Isotonic (1); Concentration equals that inside (1); No net water movement (1).	1+1+1
	OR: Water enters cells by osmosis (1); Vacuoles swell, cells become turgid (1); Stem stiffens (1).	1+1+1
18	Secretes substances (1); Gland cells (1); Salivary or pancreatic cells (1).	1+1+1
	OR: Nucleus gives instructions (1); Ribosomes on rough ER make it (1); Golgi packs; vesicle releases; mitochondria supply energy (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Mount in water and observe (1); Add strong salt or sugar solution (1); Protoplasm shrinks from the wall (1); Put back in water: recovery (1); Conclusion: the membrane is selectively permeable (1).	1+1+1+1+1
	OR: Absorption by roots (1); Turgidity and support (1); Stomata guard cells (1); Wilting (1); Preservation by salt (1).	1+1+1+1+1
20	One cell divides to form two identical daughter cells (1); Chromosome number stays the same (1); Helps growth, repair and replacement of cells (1); Meiosis gives four cells with half the chromosome number (1); Meiosis forms gametes (1).	1+1+1+1+1
	OR: RBC: no nucleus (1); Nerve: long (1); Muscle: many mitochondria (1); Sperm: tail and mitochondria (1); WBC: engulfs germs (1).	1+1+1+1+1

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